Universidad de Guadalajara
Centro Universitario de Ciencias Económico-Administrativas
Maestría en Ciencias de los Datos | 2° Semestre
Materia: Almacenamiento de Grandes Bases de Datos

Actividad: Importar Asegurados IMSS a HDFS y Hive

IMSS Asegurados 2025 – Primer Semestre – Tablas Externas en Hive

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Fecha:	6 de Abril de 2026
Entorno:	Linux Mint – Oracle VirtualBox – Hadoop 3.3.6 – YARN Single Node
Dataset:	ASG 2025 (27713294 lineas)
Fuente:	Instituto Mexicano del Seguro Social (IMSS)

1. Descripción del Dataset

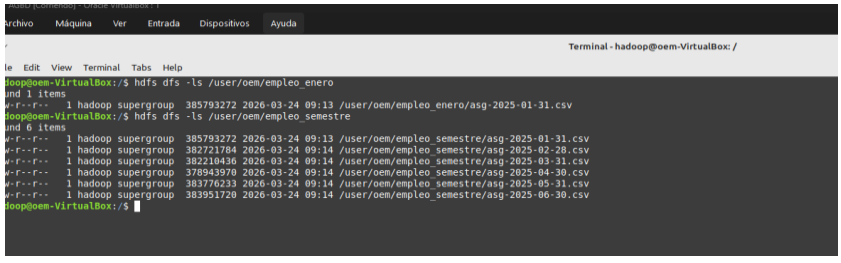
Se utilizaron los archivos de asegurados mensuales del IMSS del primer semestre de 2025, descargados del portal de datos abiertos (datos.imss.gob.mx). Cada archivo corresponde a un mes y contiene aproximadamente 4.6 millones de registros desagregados por delegación, entidad, municipio, sector económico, sexo, rango de edad, rango salarial y tipo de trabajador. El separador de campos es "|" (pipe). El semestre completo suma 27,713,294 registros.

2. Creación de Directorios en HDFS

Se crearon dos directorios en HDFS y se cargaron los archivos correspondientes:

```
# Directorio para enero 2025
hdfs dfs -mkdir -p /user/oem/empleo_enero
hdfs dfs -put /tmp/asg-2025-01-31.csv /user/oem/empleo_enero/

# Directorio para el semestre completo (6 archivos)
hdfs dfs -mkdir -p /user/oem/empleo_semestre
hdfs dfs -put /tmp/asg-2025-0*.csv /user/oem/empleo_semestre/
```



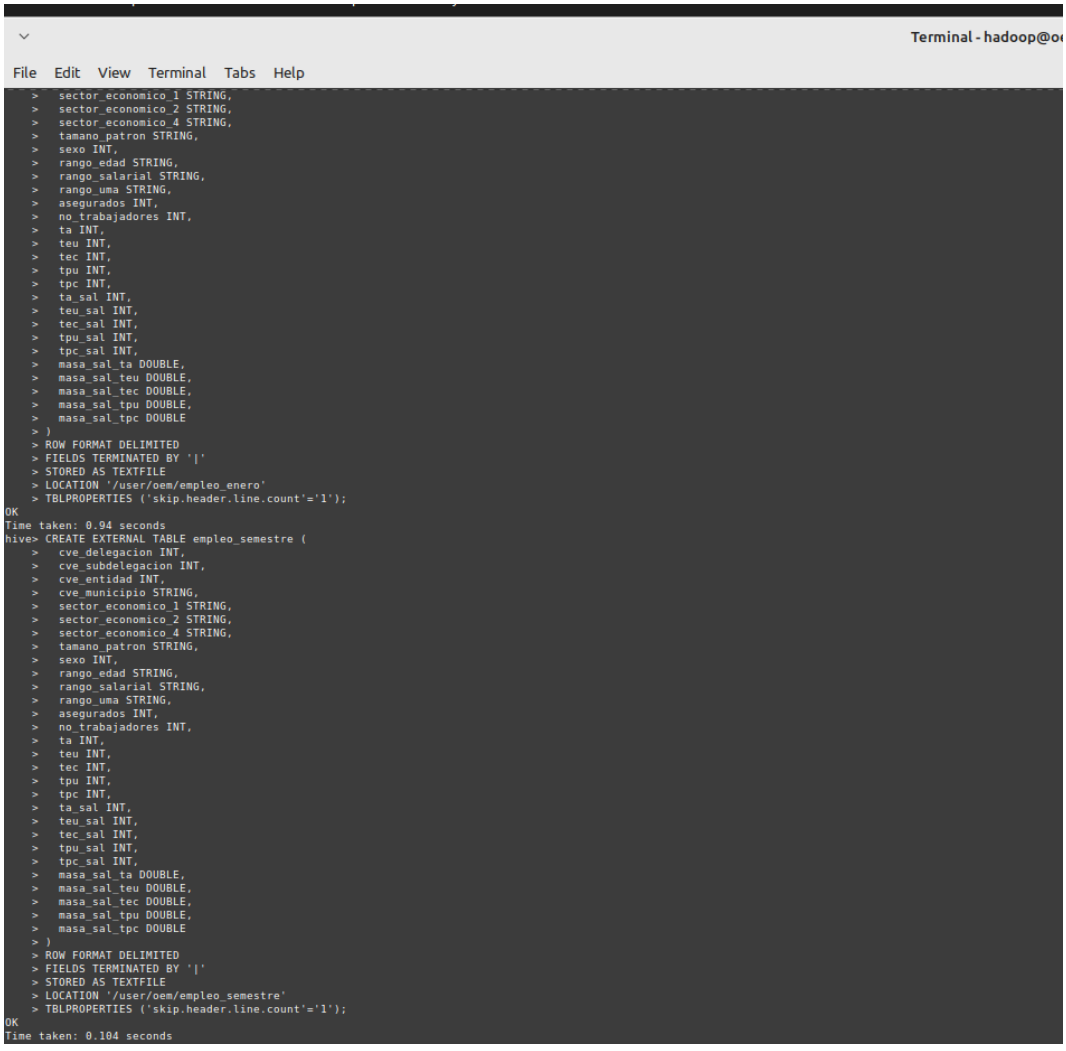
[Captura 1 – hdfs dfs -ls /user/oem/empleo_enero/ && hdfs dfs -ls /user/oem/empleo_semstre/ (7 archivos)]

3. Creación de Tablas Externas en Hive

Se crearon dos tablas externas en Hive, una por cada directorio. Una tabla externa en Hive apunta a los datos en HDFS sin moverlos — si se elimina la tabla, los archivos permanecen intactos en HDFS.

```
CREATE EXTERNAL TABLE empleo_enero (
  cve_delegacion INT, cve_subdelegacion INT, cve_entidad INT,
  cve_municipio STRING, sector_economico_1 STRING,
  sector_economico_2 STRING, sector_economico_4 STRING,
  tamano_patron STRING, sexo INT, rango_edad STRING,
  rango_salarial STRING, rango_uma STRING,
  asegurados INT, no_trabajadores INT,
  ta INT, teu INT, tec INT, tpu INT, tpc INT,
  ta_sal INT, teu_sal INT, tec_sal INT, tpu_sal INT, tpc_sal INT,
  masa_sal_ta DOUBLE, masa_sal_teu DOUBLE, masa_sal_tec DOUBLE,
  masa_sal_tpu DOUBLE, masa_sal_tpc DOUBLE
) ROW FORMAT DELIMITED FIELDS TERMINATED BY '|'
STORED AS TEXTFILE LOCATION '/user/oem/empleo_enero'
TBLPROPERTIES ('skip.header.line.count'='1');

CREATE EXTERNAL TABLE empleo_semestre ( ... mismas columnas ... )
STORED AS TEXTFILE LOCATION '/user/oem/empleo_semestre'
TBLPROPERTIES ('skip.header.line.count'='1');
```

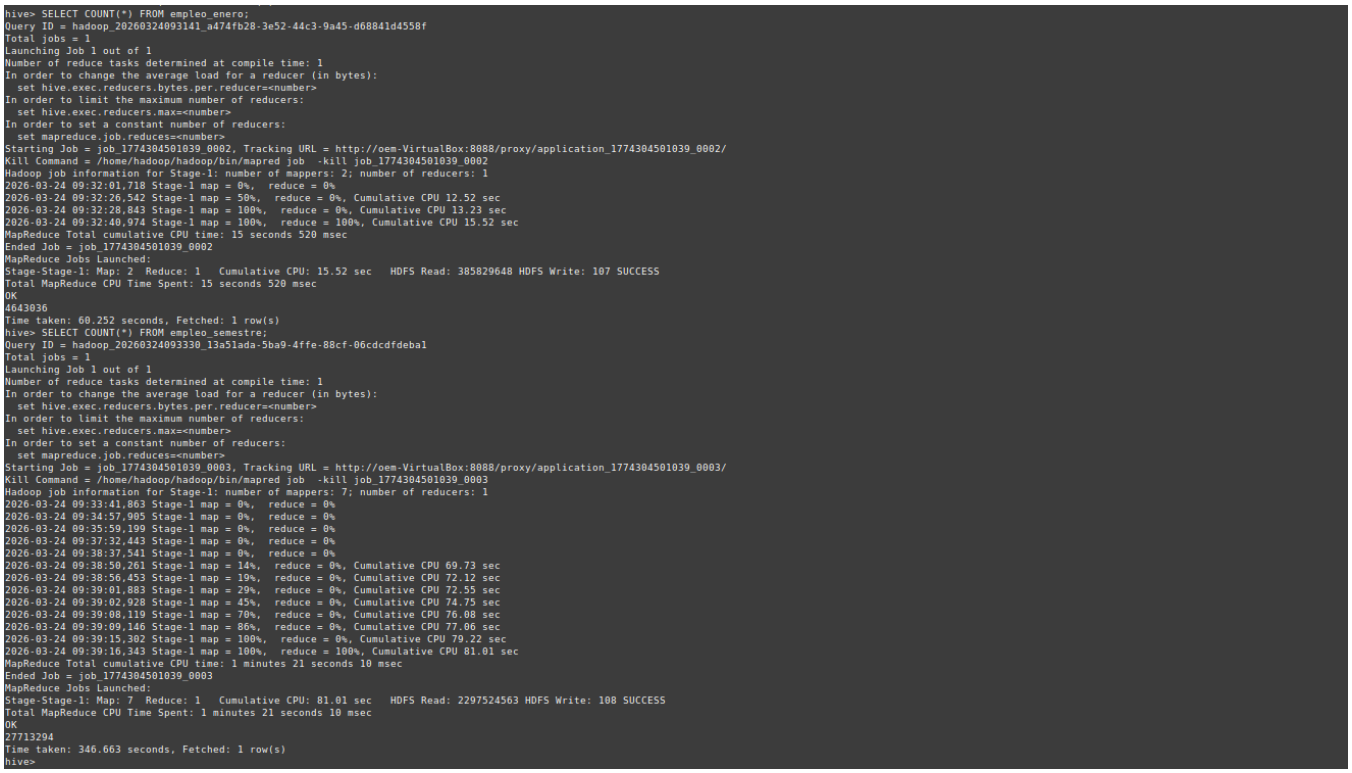


[Captura 2 – SHOW TABLES mostrando empleo_enero y empleo_semestre]

4. Verificación con Consultas SQL

Se ejecutaron consultas COUNT(*) para verificar la carga correcta de los datos en ambas tablas:

```
SELECT COUNT(*) FROM empleo_enero;     -- Resultado: 4,643,036
SELECT COUNT(*) FROM empleo_semestre;  -- Resultado: 27,713,294
```



```
hive> SELECT COUNT(*) FROM empleo_enero;
Query ID = hadoop_20260324093141_a474fb28-3e52-44c3-9a45-d68841d4558f
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1774304501039_0002, Tracking URL = http://oem-VirtualBox:8088/proxy/application_1774304501039_0002/
Kill Command = /home/hadoop/hadoop/bin/mapred job  -kill job_1774304501039_0002
Hadoop job information for Stage-1: number of mappers: 2; number of reducers: 1
2026-03-24 09:32:01,718 Stage-1 map = 0%,  reduce = 0%
2026-03-24 09:32:26,542 Stage-1 map = 50%,  reduce = 0%, Cumulative CPU 12.52 sec
2026-03-24 09:32:28,843 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 13.23 sec
2026-03-24 09:32:40,974 Stage-1 map = 100%,  reduce = 100%, Cumulative CPU 15.52 sec
MapReduce Total cumulative CPU time: 15 seconds 520 msec
Ended Job = job_1774304501039_0002
MapReduce Jobs Launched:
Stage-Stage-1: Map: 2  Reduce: 1   Cumulative CPU: 15.52 sec   HDFS Read: 385829648 HDFS Write: 107 SUCCESS
Total MapReduce CPU Time Spent: 15 seconds 520 msec
OK
4643036
Time taken: 60.252 seconds, Fetched: 1 row(s)
hive> SELECT COUNT(*) FROM empleo_semestre;
Query ID = hadoop_20260324093330_13a51ada-5ba9-4ffe-88cf-06cdcdfdeba1
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1774304501039_0003, Tracking URL = http://oem-VirtualBox:8088/proxy/application_1774304501039_0003/
Kill Command = /home/hadoop/hadoop/bin/mapred job  -kill job_1774304501039_0003
Hadoop job information for Stage-1: number of mappers: 7; number of reducers: 1
2026-03-24 09:33:41,863 Stage-1 map = 0%,  reduce = 0%
2026-03-24 09:34:57,905 Stage-1 map = 0%,  reduce = 0%
2026-03-24 09:35:59,199 Stage-1 map = 0%,  reduce = 0%
2026-03-24 09:37:32,443 Stage-1 map = 0%,  reduce = 0%
2026-03-24 09:38:37,541 Stage-1 map = 0%,  reduce = 0%
2026-03-24 09:38:50,261 Stage-1 map = 14%,  reduce = 0%, Cumulative CPU 69.73 sec
2026-03-24 09:38:56,453 Stage-1 map = 19%,  reduce = 0%, Cumulative CPU 72.12 sec
2026-03-24 09:39:01,883 Stage-1 map = 29%,  reduce = 0%, Cumulative CPU 72.55 sec
2026-03-24 09:39:02,928 Stage-1 map = 45%,  reduce = 0%, Cumulative CPU 74.75 sec
2026-03-24 09:39:08,119 Stage-1 map = 70%,  reduce = 0%, Cumulative CPU 76.08 sec
2026-03-24 09:39:09,146 Stage-1 map = 86%,  reduce = 0%, Cumulative CPU 77.06 sec
2026-03-24 09:39:15,302 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 79.22 sec
2026-03-24 09:39:16,343 Stage-1 map = 100%,  reduce = 100%, Cumulative CPU 81.01 sec
MapReduce Total cumulative CPU time: 1 minutes 21 seconds 10 msec
Ended Job = job_1774304501039_0003
MapReduce Jobs Launched:
Stage-Stage-1: Map: 7  Reduce: 1   Cumulative CPU: 81.01 sec   HDFS Read: 2297524563 HDFS Write: 108 SUCCESS
Total MapReduce CPU Time Spent: 1 minutes 21 seconds 10 msec
OK
27713294
Time taken: 346.663 seconds, Fetched: 1 row(s)
hive>
```

[Captura 3 – COUNT(*) de las tablas]